MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

PRESENTATION ON

RIP: Routing Information Protocol

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Course: **Data Communication**

Course Code: **ICT-3101**

Topic: RIP

Presentation roadmap

1

Foundations 01

Why routers need a dynamic routing protocol

2

RIP in action 02

Distance vectors, hop counts and route updates

3

Versions and resilience 03

RIPv1, RIPv2, RIPng and loop prevention

4

Evaluation 04

Advantages, limitations and protocol comparison

The routing problem

Many networks

Routers must discover paths to destinations beyond .directly connected links

Networks change

Links can fail, new networks appear and manual updates .become difficult

A shared solution

Dynamic routing lets routers exchange reachability .information

?What is RIP

1

Distance-vector routing protocol

.Routers advertise destination networks and their distance in hops

2

Bellman–Ford principle

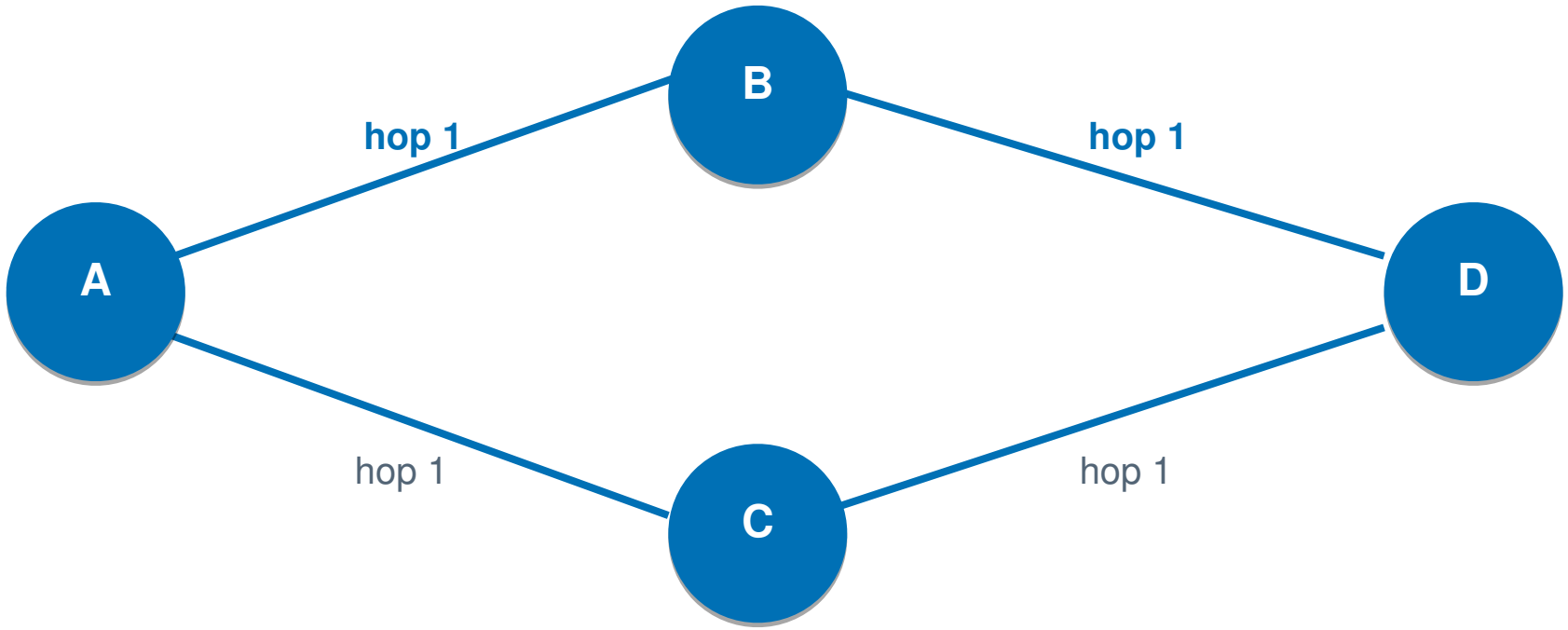
.Each router estimates the best route using information from neighbours

3

Interior gateway protocol

.Designed for routing within a single administrative domain

RIP topology | Finding the shortest path



A → D

via B: 2 hops

via C: 2 hops

.Illustrative topology: RIP compares hop count, not bandwidth or delay

How RIP learns a route

1

Step 1 | Discover

.Each router knows its directly connected networks

2

Step 2 | Advertise

.Neighbours exchange routing entries, typically every 30 seconds

3

Step 3 | Update

.A router adds one hop and selects a route with the lowest hop count

4

Step 4 | Adapt

.Changed or unreachable routes trigger routing-table updates

A simple routing-table example

Destination	Next hop	Hop count
192.168.10.0/24	Direct	0
192.168.20.0/24	Router B	1
192.168.30.0/24	Router B	2
192.168.40.0/24	Router C	3

.Illustrative entries. Connected networks are shown as 0 hops locally

RIP essentials

Metric

Hop count only; the route with the fewest hops is .preferred

Maximum reach

hops is the 15 maximum usable metric; 16 means .unreachable

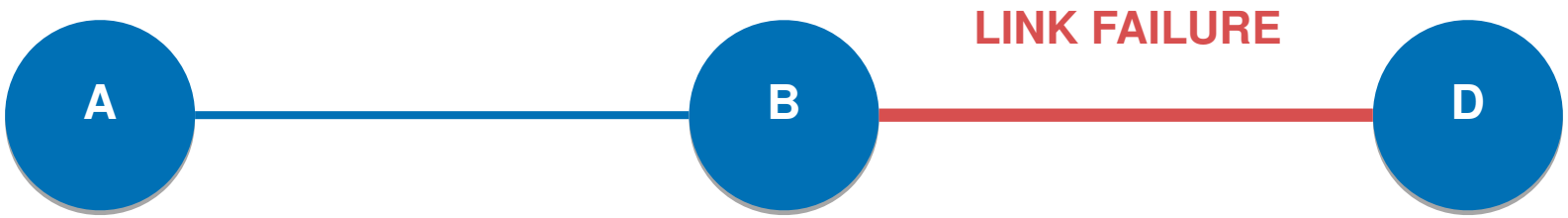
Updates

Periodic routing advertisements, normally every 30 .seconds

Transport

RIPv1 and RIPv2 .use UDP port 520

When a link fails | Convergence



Route becomes unavailable 1

Router advertises the change 2

Neighbours recalculate their tables 3

.RIP convergence can be slow, especially in larger or unstable networks

How RIP reduces routing loops

Split horizon

Do not advertise a route back through the interface .from which it was learned

Route poisoning

Advertise a failed destination .with metric 16 (unreachable)

Triggered updates

Send an update promptly after a significant routing .change

RIP versions | At a glance

Feature	RIPv1	RIPv2	RIPng
Address family	IPv4	IPv4	IPv6
Classless / CIDR	No	Yes	Yes
Update delivery	Broadcast	Multicast	Multicast
Authentication	No	Supported	Uses IPv6 security mechanisms
UDP port	520	520	521

.RIPng is the IPv6 adaptation of RIP; it is not simply RIPv2 with longer addresses

Advantages and limitations

1

Why RIP is still useful

.Easy to understand, simple to configure and appropriate for small networks or labs

2

Where RIP struggles

.hop limit, slow convergence and no bandwidth-aware path selection-15

3

Practical takeaway

.A straightforward teaching protocol, but often not the first choice for large modern networks

RIP vs OSPF vs EIGRP

Aspect	RIP	OSPF	EIGRP
Approach	Distance vector	Link state	Advanced distance vector
Main metric	Hop count	Cost	Composite metric
Scalability	Small networks	Large networks	Medium to large
Convergence	Relatively slow	Generally fast	Generally fast
Complexity	Low	Higher	Moderate

.Performance depends on topology, implementation and configuration

Where RIP makes sense

Networking labs

Learn routing tables, advertisements, metrics and .convergence

Small legacy networks

Simple environments with limited size and modest .routing requirements

Protocol fundamentals

A clear introduction before studying OSPF, IS-IS or .BGP

Key takeaways

1

RIP exchanges distance vectors

.Routers learn routes through advertisements from their neighbours

2

Hop count drives decisions

.hops is the maximum usable distance; 16 means unreachable 15

3

Simplicity has trade-offs

.RIP is accessible but less scalable and slower to converge than modern alternatives

Selected references

.RFC 1058 — Routing Information Protocol (RIPv1), C. Hedrick, 1988 [1]

.RFC 2453 — RIP Version 2, G. Malkin, 1998 [2]

.RFC 2080 — RIPng for IPv6, G. Malkin and R. Minnear, 1997 [3]

.RFC 4822 — RIPv2 Cryptographic Authentication, R. Atkinson and M. Fanto, 2007 [4]

THANK YOU